

AirWise: LLM-Powered Smart Air Quality Monitoring and Automated Appliance Control

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Abstract—Indoor air quality (IAQ) critically affects occupant health, comfort, and energy efficiency in smart homes. Most current IAQ solutions focus on pollutant detection and visualization, offering limited support for adaptive, user-specific appliance control. This paper presents AirWise, a large language model (LLM)-based framework that integrates real-time indoor and outdoor environmental data with user profiles, including health conditions, comfort preferences, and appliance availability, to provide personalized, context-aware control recommendations. AirWise leverages the Environmental Monitoring Device (EMD) on the iAirGuard platform, which ensures high-quality input by denoising sensor data and using TinyML-based fault detection. The LLM functions as a high-level decision engine, dynamically adjusting AC mode, fan speed, and window/door states, balancing pollutant mitigation, comfort, and energy efficiency. The framework was evaluated in a real smart home testbed under severe pollution conditions. During experiments with an AC equipped with a filter, indoor PM_{2.5} and PM₁₀ AQI dropped from 6 (Severe) to 2 (Satisfactory) within 50 minutes. LLM inference consumed less than 5% of system tokens, demonstrating real-time applicability. AirWise offers a scalable, adaptive, and energy-aware solution for personalized IAQ management.

Index Terms—Indoor air quality (IAQ), Smart homes, Large language models (LLMs), TinyML, IoT sensing, Automated appliance control, Energy-efficient personalized comfort

I. INTRODUCTION

Indoor air quality (IAQ) has a substantial impact on human health, comfort, and energy consumption in residential buildings. It is estimated that indoor dwellers are exposed to various sources of pollution like PM_{2.5}, PM₁₀, VOCs, and CO₂, and temperature and humidity fluctuations that may affect their health through respiratory diseases, allergies, reduced cognition, and general discomfort [1], [2]. Efficient and health-supportive IAQ management requires a steady monitoring system, along with the control of appliances, to maintain comfort while minimizing energy consumption [3], [4].

Present-day systems primarily focus on detecting pollutants, visualising them, and, in some cases, utilizing machine learning for predictive modelling [2], [5]. Nevertheless, the most common practice is for them to run separately, neglecting

user preference, health conditions, appliance availability, and the changing weather conditions outside. Making real-time, personalized decisions with heterogeneous data remains a challenge. Although recent studies have examined the use of Large Language Models (LLMs) in smart building support [6]–[8], their application for combining multi-objective, IAQ-aware appliance control remains an open research problem.

To fill these gaps, we introduce **AirWise**, a platform powered by an LLM that monitors indoor air quality in real time and automates appliance control. AirWise connects the dots between environmental data from indoors and outdoors, appliance operation logs for every device, and user profiles corresponding to them, to craft control strategies that are not only context-aware but also personalized. The LLM reasoning engine of AirWise constantly adjusts AC modes, fan speeds, and even window/door states to effectively reduce pollutants, keep occupants comfortable, and simultaneously consume less energy. The **Environmental Monitoring Device (EMD)** from the iAirGuard platform is responsible for providing the input that is trustworthy; it utilizes TinyML-based sensor denoising and fault detection methods [9]. The main contributions of this work are as follows:

- **Integrated IAQ Monitoring and Control:** Combination of indoor/outdoor sensing, appliance data, and user preferences for managing IAQ in real time.
- **LLM-Based Personalized Reasoning:** Multi-objective, context-aware recommendations for air quality, comfort, health, and energy efficiency.
- **Experimental Validation:** Real smart home case studies show significant improvements in IAQ, comfort, and energy use.
- **Scalable, Occupant-Centric Design:** An adjustable system that caters to a range of residential settings and environmental conditions.

The rest of the paper is organized as follows. Section 2 reviews the existing literature on IAQ management, smart homes, and building control using LLMs. Section 3 presents the system

architecture of AirWise. Section 4 describes the methodology, outlining sensing, user profiling, IoT communication, and LLM-based recommendations. Section 5 discusses the experimental setup and presents three case studies that illustrate the system's performance. Finally, Section 6 concludes the paper and gives some future directions.

II. RELATED WORK

The indoor air quality (IAQ) monitoring has been widely studied through the use of IoT-based sensor networks that are capable of measuring the pollutants, such as PM_{2.5}, PM₁₀, VOCs, and CO₂, as well as the environmental parameters of temperature and humidity continuously [1], [3], [10]. It enables the monitoring of conditions and processes, as well as real-time visualization, establishing the possibility of utilizing such systems in residential and commercial areas. Nevertheless, most of the current methods primarily focus on acquiring and reporting data, thereby providing limited support for adaptive device control or personalization based on user preferences, health status, or energy efficiency.

The concepts of machine learning (ML) and artificial intelligence (AI) have been utilized to predict IAQ trends, as well as their subsequent roles in the decision-making process. A combination of hybrid deep learning models and metaheuristic optimization algorithms has been utilized to predict indoor pollutant levels and to determine the optimal HVAC(...) operation [2], [5]. In particular, deep learning methods are applied to improve indoor environmental quality in smart buildings [8]. However, the methods mentioned above increase prediction accuracy and allow for partial automation, they still extensively rely on narrow input feature definition, separate models for different objectives, and a lack of incorporation of the real-time, user-specific context or qualitative comfort preferences. Furthermore, the issue of sensor data reliability, which is of great importance for AI-based control, is not thoroughly studied in these works.

Recently, the advent of Large Language Models (LLMs) has introduced a new dimension of reasoning capabilities in smart buildings. Previous studies have reported on LLM-based assistants and decision-support systems capable of merging and analyzing data from various sources to provide context-aware recommendations [6], [7]. These studies showcase the remarkable ability of LLMs to link sensing, reasoning, and user interaction. On the other hand, these studies are still quite limited to natural language interfaces and high-level building operations, thereby ignoring user-level personalization and real-time IAQ management, especially in situations where sensors provide noisy or unreliable data.

Synthesis and Research Gap

In summary, IoT sensors provide the basis for accurate monitoring, ML provides the basis for predictive control, and LLMs provide the basis for flexible reasoning. However, existing solutions mainly consider these components in isolation. Only a handful of methods bring together the three processes - *reliable edge-level detection*, *multi-source contextual reasoning*, and *dynamic appliance custody* at the

same time, talking about occupant health, comfort preferences, and energy efficiency.

Problem Statement

Current IAQ systems lack a unified framework for adaptive, personalized, and energy-efficient indoor air quality management. The majority focus solely on environmental monitoring or fail to disseminate control strategies for various users and conditions. The combination of IoT-based sensing, TinyML-improved data reliability, and LLM-based reasoning has not yet received the attention it deserves; therefore, there is a need to develop a user-centric, smart, and intelligent home solution.

III. SYSTEM ARCHITECTURE

The AirWise system combines reliable environmental sensing with intelligent LLM-guided reasoning to dynamically manage indoor air quality (IAQ). It consists of two primary parts: (i) **iAirGuard**, the edge-processing sensing lower layer, and (ii) the **AirWise framework** that handles context-aware reasoning and appliance-level control (Fig. 2).

A. iAirGuard: Preliminary Sensing and Edge Processing Framework

iAirGuard is the basic sensing layer of AirWise. It uses an **Environmental Monitoring Device (EMD)** Fig. 1a that comes with the sensors Fig. 1b for PM_{2.5}, PM₁₀, VOCs, CO₂, temperature, and humidity, thus allowing very detailed continuous indoor monitoring. The tinyML-based edge pre-processing suppresses signal noise, detects sensor faults and failures, and enhances the quality of the readings sent, thereby reducing communication overhead, preventing incorrect data from reaching the LLM, and so on. It also provides a reliable power-

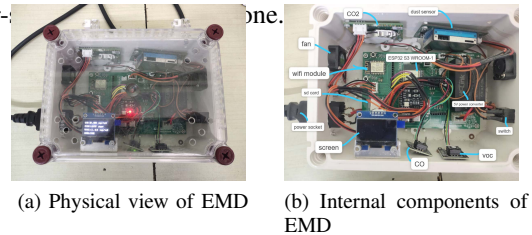


Fig. 1. iAirGuard Environmental Monitoring Device (EMD) showing (a) physical deployment and (b) internal sensing and edge-processing components.

Through the reliable detection of iAirGuard [9], the AirWise system measures indoor and outdoor air quality, appliance states, and user preferences regarding comfort and health. The LLM-based reasoning module subsequently processes these ubiquitous data to formulate control strategies that are a) flexible and suitable for multiple objectives, such as pollutant reduction, comfort, and energy efficiency, at the same time.

AirWise operates devices through A/C modes and temperatures, fan speed controls, and window/door operation. A centralized feedback loop is in place that continually monitors indoor conditions, allowing recommendations to be updated dynamically and instantly. In contrast to traditional systems based on rules or machine learning, LLMs combine multiple objectives with qualitative context. The flexible feedback-driven structure allows for rapid expansion to cover various

types of housing with the same speed, dependability, and people-centric IAQ management (Fig. 2).

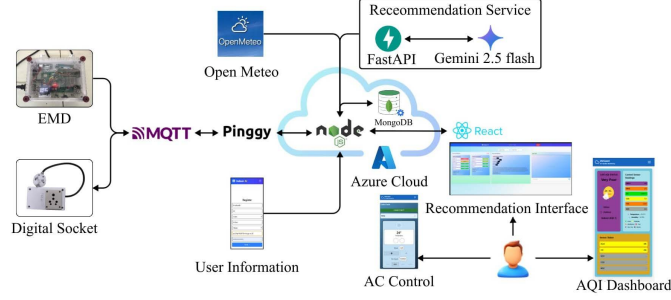


Fig. 2. AirWise system architecture.

IV. METHODOLOGY

The AirWise framework is a closed-loop IAQ management system that integrates reliable IoT sensing, secure data communication, user-centric modelling, and LLM-driven reasoning to optimise comfort, reduce pollutants, and conserve energy, adapting dynamically to changing conditions and user feedback. An overview of the methodological workflow is illustrated in Fig. 3.

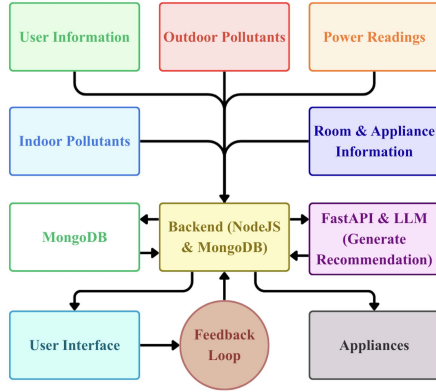


Fig. 3. Overall architecture and methodological workflow of the proposed AirWise framework.

A. Indoor Environmental Sensing and Edge Preprocessing

Indoor air quality and comfort parameters ($PM_{2.5}$, PM_{10} , CO_2 , VOCs, temperature, humidity) are continuously measured using the iAirGuard EMD. TinyML-based edge processing performs denoising, anomaly detection, and fault identification, ensuring reliable data and reducing communication overhead before transmission to the backend, as shown in Fig. 4.

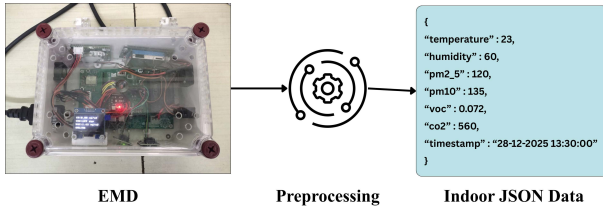


Fig. 4. EMD with its JSON data.

B. Outdoor Environmental Data Acquisition

Outdoor air quality and weather data are obtained in real time via the Open-Meteo API, including $PM_{2.5}$, PM_{10} , CO , dust, temperature, humidity, wind, precipitation, and day/night status, as illustrated in Fig. 5. Derived metadata, such as season and time of day, provide context for ventilation and appliance control, helping balance natural ventilation against outdoor pollution.

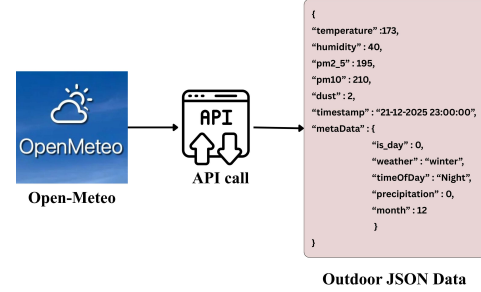


Fig. 5. Outdoor JSON data with OpenMeteo.

C. Appliance-Level Power Monitoring and Control

Energy-aware control is achieved through a custom Digital Socket featuring an ESP32-S3 Nano, relay switching, a current sensor, and an IR module. It continuously monitors RMS current, power, and energy consumption, while autonomously actuating the AC using IR pwm signal pre coded in the socket, based on the received commands, enabling real-time adjustment of modes and temperature, as shown in Fig. 6.

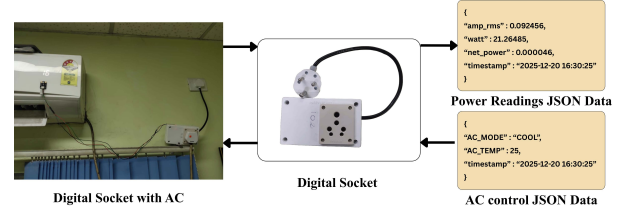


Fig. 6. Power readings JSON data with Digital Socket.

D. Secure IoT Communication and Data Management

As Fig. 7 shows IoT devices, including the EMD and Digital Socket, communicate via the lightweight MQTT protocol, secured using Pinggy.io tunnels. A cloud backend on Microsoft Azure processes and stores incoming data, while MongoDB maintains time-series records of sensor readings, appliance logs, recommendations, and user feedback. This ensures secure, reliable communication, facilitates data-driven personalization, and supports longitudinal studies for indoor air quality management and system optimization.



Fig. 7. Secure IoT communication through Pinggy and MQTT.

E. User Profiling and Preference Acquisition

User information is collected through a web interface during onboarding, including demographics, health information, comfort preferences, and ventilation habits, as shown in Fig. 8. A 21-question survey captures detailed IAQ and comfort needs. All data is stored in MongoDB and referenced continuously to guide personalized LLM-based recommendations.

Fig. 8. User preferences webpage preview.

F. Data Aggregation and Backend Processing

A Node.js backend developed based on the Express.js framework combines various data streams, including verified indoor sensor readings, outdoor environmental data, appliance power usage, user profiles, room setup details (size, number of people, and available appliances), and previous recommendations. The combined data is then structured and sent to a FastAPI-based service, which acts as a mediator between the backend system and the LLM-based reasoning engine.

G. LLM-Based Reasoning and Recommendation Generation

AirWise uses the **Gemini 2.5 Flash LLM** via cloud API to perform real-time, context-aware IAQ management. The LLM synthesizes indoor/outdoor environmental data, user health and comfort profiles, room and appliance states, power metrics, and historical feedback to generate structured recommendations. One-shot prompting with a single input-output example ensures responses follow a strict JSON schema, reducing token usage and latency while maintaining reasoning accuracy.

(i) *Gemini 2.5 Flash API Usage and Token Details:* AirWise uses the **Gemini 2.5 Flash LLM** with a **1,048,576-token context window** and up to **65,536 output tokens**. Typical usage includes ~1,800- 2,000 prompt tokens (sensor data, user profiles, one-shot example), ~2,500- 3,000 internal reasoning tokens, and ~306- 400 output tokens per recommendation.

(ii) *Token::* A unit of text (word, punctuation, or subword) processed by the LLM. For example, “AirWise manages PM_{2.5}” tokenizes to 6-8 tokens.

(iii) *Structured One-Shot Prompt:* The structured prompt sent to Gemini 2.5 Flash includes the following components, as illustrated in Fig. 9:

- 1) Indoor and outdoor environmental readings (validated PM_{2.5}/PM₁₀/CO₂/VOCs, temperature, humidity)
- 2) User profile data (health conditions, comfort preferences, seasonal sensitivity)
- 3) Room configuration and appliance availability

- 4) Historical recommendations and feedback (if available)
- 5) A single example demonstrating correct JSON-formatted appliance control output

```
You are an advanced **Indoor Environmental Comfort & Air Quality AI Assistant**.
Your goal is to provide *personalized appliance recommendations* to optimize
comfort, air quality, and energy efficiency — based on **indoor/outdoor
conditions**,
**user feedback**, and **room configuration**.

...

### USER INFORMATION
- Name: Prashanth

...
```

One-Shot Prompt

Fig. 9. Abstracted structure of the one-shot prompt.

LLM Output and Control Integration: The LLM output specifies actionable control commands, as presented in Fig. 10, including:

- AC operating mode and temperature
- Fan speed
- Window/door states
- Re-evaluation interval (RECHECK_AT)
- Concise rationale for each action

```
{
  "reason": "Indoor PM2.5 and PM10 concentrations are
significantly elevated while outdoor air quality is poor. To
reduce pollutant exposure and maintain thermal comfort,
the AC is operated in cooling mode with moderate fan
circulation while keeping windows and doors closed.
Expected improvements are anticipated within the re-
evaluation interval.",
  "AC_MODE": "COOL",
  "AC_TEMPERATURE": 23,
  "CEILING_FAN": 2,
  "WINDOW": "CLOSED",
  "DOOR": "CLOSED",
  "RECHECK_AT": 15
}
```

Recommendation Output JSON Data

Fig. 10. An output Example JSON-formatted appliance control recommendation.

By leveraging one-shot prompting with Gemini 2.5 Flash, AirWise achieves **real-time, personalized IAQ recommendations** with minimal token usage (less than 5% of total system tokens per inference), ensuring practical and energy-efficient deployment in residential smart home environments.

H. Automated Actuation and Closed-Loop Feedback

LLM-generated suggestions are stored in the database, displayed on the real-time web interface, and sent via MQTT to the Digital Socket for automated AC control. Environmental conditions are continuously monitored post-actuation. A dynamic re-evaluation interval (RECHECK_AT) is computed based on IAQ severity. At each interval, users provide comfort

feedback, which, along with override actions and conversational inputs, is incorporated into subsequent LLM prompts, enabling adaptive human-in-the-loop control, as depicted in Fig. 11.

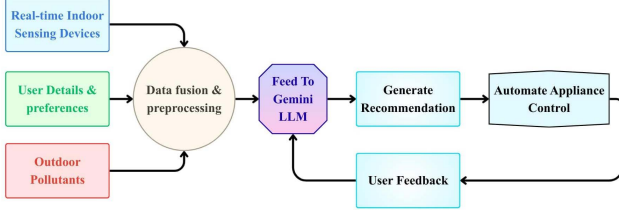


Fig. 11. LLM-driven reasoning and closed-loop control in AirWise.

I. User Interface and Conversational Interaction

The AirWise user interface consists of three main parts: (i) a real-time IAQ dashboard with colour-coded severity levels, (ii) a recommendation panel showing appliance settings, reasons, and time for re-evaluation, and (iii) a context-aware conversational assistant powered by Gemini 2.5 Flash, as shown in Fig. 12.

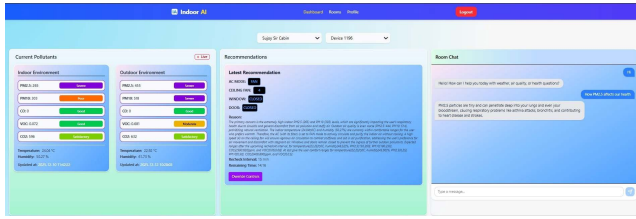


Fig. 12. AirWise user interface overview.

The conversational module is limited to IAQ, health, and environment-related conversations and can identify user intent (e.g., thermal discomfort) to start the re-optimization of suggestions. Additionally, remote appliance control is supported, enabling users to manage their devices from anywhere via secure MQTT communication.

V. EXPERIMENTAL SETUP AND CASE STUDIES

A. Experimental Setup

The experiments were carried out on a smart home testbed equipped with sensors for $PM_{2.5}$, PM_{10} , CO_2 , VOCs, temperature, and humidity, in addition to controllable appliances including an AC (with/without filter), a ceiling fan, and windows/doors. Sensor reliability was ensured through iAirGuard [9] EMD-based fault detection. Indoor/outdoor air quality data was collected continuously, and an LLM-based reasoning module generated dynamic appliance control recommendations to optimize comfort, reduce pollutants, and minimise energy use. Controlled cigarette-burning scenarios simulated pollutant spikes, with indoor AQI monitored on a 1–6 scale (1: Good, 6: Severe).

B. Case Study 1: AC with Filter, User with Sinusitis

Conducted on 8th November 2025 (winter), the user had sinusitis. The initial $PM_{2.5}/PM_{10}$ AQI was 5/3. Cigarette burning from 15:15–15:31 raised PM levels to AQI 6 ($>400 \mu g/m^3$). The first recommendation was to set the AC to Fan mode, set the ceiling fan to level 3, and open the windows/doors

for 15 minutes. This minimized energy usage (avoiding AC cooling), leveraged the AC filter to reduce pollutants, and allowed natural ventilation as indoor AQI exceeded outdoor AQI. After 15 minutes, $PM_{2.5}/PM_{10}$ AQI dropped to 4/3, as shown in Fig. 13a and Fig. 13b.

The second recommendation kept the AC in Fan mode, at level 2, and closed windows/doors for 15 minutes, as the indoor and outdoor AQIs were comparable, reducing $PM_{2.5}/PM_{10}$ to 3/2. The third recommendation maintained AC in Fan mode, with the fan off, and windows/doors closed for 20 minutes due to a high outdoor AQI, further lowering $PM_{2.5}/PM_{10}$ to 2/1, as depicted in Fig. 13.

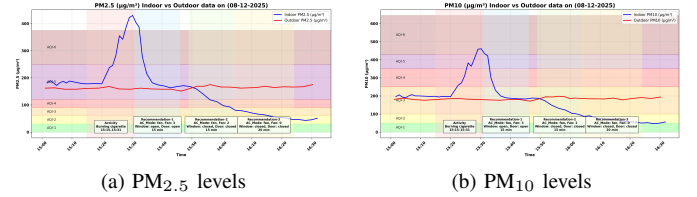


Fig. 13. Case Study 1: Indoor and outdoor pollutant levels with recommendation periods shaded.

C. Case Study 2: AC without Filter, User with Sinusitis and Asthma

Performed on 9th November 2025, the user had sinusitis and asthma, and the AC had no filter. Initial $PM_{2.5}/PM_{10}$ AQI were 6/5. Cigarette burning from 11:30–11:42 increased PM to AQI 6. The first recommendation kept AC off, set the fan to level 3, and closed the windows and doors to prevent outdoor infiltration. After 15 minutes, $PM_{2.5}/PM_{10}$ remained high (6/6), as shown in Fig. 14a and Fig. 14b.

The second recommendation opened windows/doors and set the fan level to 3 for 15 minutes, allowing natural ventilation due to slightly cleaner outdoor air, which reduced $PM_{2.5}/PM_{10}$ to $360/380 \mu g/m^3$. Finally, windows/doors closed, fan level 2, improving comfort while further lowering $PM_{2.5}/PM_{10}$ to $290/320 \mu g/m^3$, closely matching outdoor levels, as shown in Fig. 14.

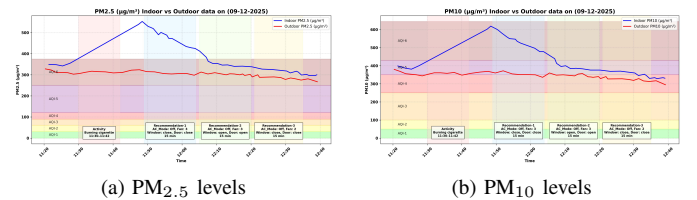


Fig. 14. Case Study 2: Indoor and outdoor pollutant levels with recommendation periods shaded.

D. Case Study 3: No AC, User without Health Issues

Also on 9th November 2025, a healthy user without AC. Initial $PM_{2.5}/PM_{10}$ AQI were 6/4. Cigarette burning from 13:00–13:12 raised AQI to 6. The first recommendation was to set the ceiling fan to level 3, open windows/doors for 15 minutes, and leverage outdoor ventilation to reduce pollutants, as shown in Fig. 15a and Fig. 15b. The second recommendation is to close windows/doors, and set the fan

to level 3 for 15 minutes to stabilize the indoor AQI with outdoor air. The final recommendation reduced the fan to level 2, with windows/doors closed for 15 minutes, resulting in $\text{PM}_{2.5}/\text{PM}_{10}$ levels of $240/270 \mu\text{g}/\text{m}^3$, comparable to outdoor levels, as depicted in Fig. 15.

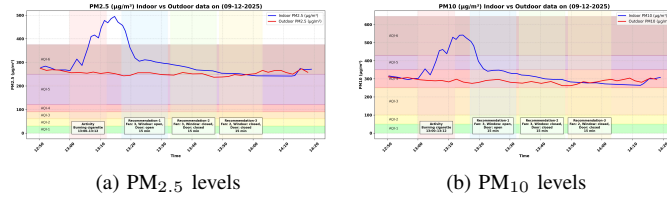


Fig. 15. Case Study 3: Indoor and outdoor pollutant levels with recommendation periods shaded.

E. Observations and Comparison

All three cases successfully controlled pollutants while maintaining comfort and energy efficiency. Case Study 1 highlighted the effectiveness of the AC filter, Case Study 2 relied on ventilation and fan circulation in the absence of a filter, and Case Study 3 utilized fan and window/door control to maintain an indoor AQI close to that of the outdoor air, as summarized in Table I.

TABLE I
SUMMARY OF INDOOR AIR QUALITY AND SYSTEM PERFORMANCE
ACROSS CASE STUDIES

Metric	Case 1	Case 2	Case 3
$\text{PM}_{2.5}$ AQI Reduction	6 $\rightarrow$ 2	6 $\rightarrow$ 6 $\rightarrow$ 6 $\rightarrow$ 5	6 $\rightarrow$ 5
PM_{10} AQI Reduction	6 $\rightarrow$ 1	5 $\rightarrow$ 4	4 $\rightarrow$ 4
Energy Usage	Minimized	Minimized	Minimized
User Comfort	High	High	High

These results confirm AirWise’s adaptability, efficiency, and ability to maintain indoor air quality across different users, health conditions, and appliance setups.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presents **AirWise**, a smart home system that leverages LLM-based reasoning over sensor data to manage comfort, reduce pollution, and improve energy efficiency. It integrates real-time indoor and outdoor environmental data with user profiles, such as health conditions, comfort preferences, and appliance availability, to give personalized, context-aware control recommendations. Case studies shown in this work demonstrate the effective control of AC, fan, and window/door operations, resulting in lowered $\text{PM}_{2.5}$ and PM_{10} levels, maintained temperature and humidity, and optimized energy use. Furthermore, TinyML-enhanced edge sensing via *iAirGuard* ensured reliable data, enabling real-time adaptive recommendations with less than 5% of the LLM token budget, highlighting the feasibility of deploying LLM-assisted decision-making in resource-constrained smart home environments. Sensor details, prompt template, and example outputs are available in our open-source repository [11].

B. Limitation and Future Work

In the present study, comparative evaluations with alternative control strategies, a systematic analysis of different LLM architectures and prompting techniques, robustness under sensor or actuator failures, and a detailed end-to-end latency assessment are not investigated and are left for future research. In addition, we plan to fully automate appliances, deploy locally hosted LLMs for privacy and low latency, explore diverse prompting strategies, and incorporate predictive, health-aware, and multi-parameter IAQ control in the future.

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